



DevOps (Cloud DevOps) - Lab 01

**GitHub Identity, Student Portal Registration, and Ubuntu Server
Installation**

Lab Number and Title	Lab 01: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation
Complete Name	Kiran Siddique
Registration Number	2023-BSCS-52
Course Code	CS-649 (Cloud DevOps)
Section	A
GitHub Username	kiransiddique



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Task 1: Create and Configure Your GitHub Account

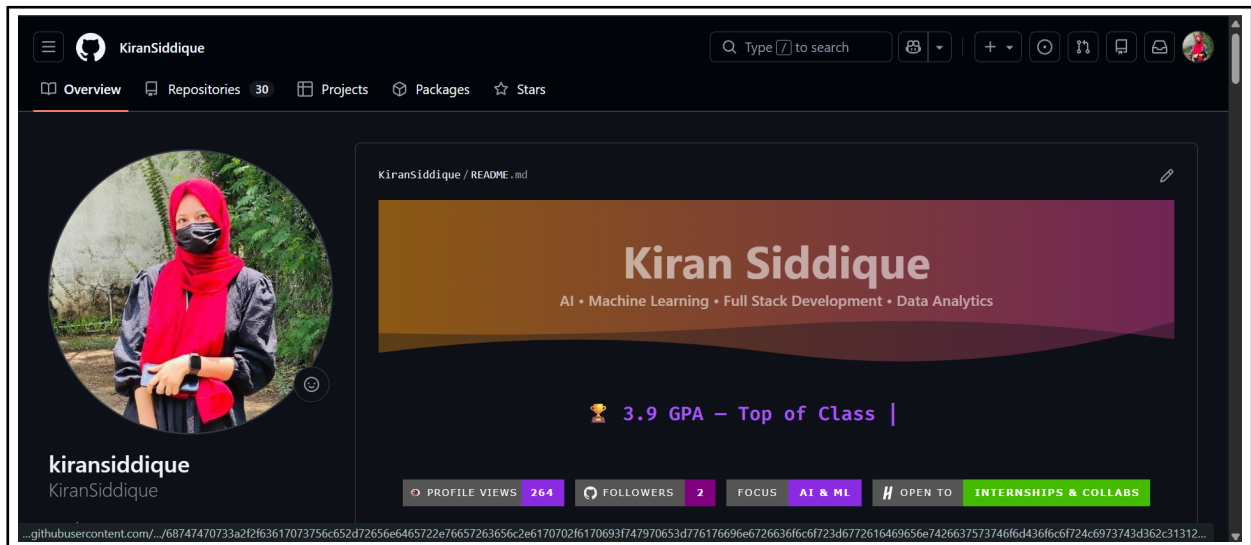


Figure: GitHub profile showing the profile name and GitHub username (kiransiddique).



Task 2: Register in the Student Marks Portal

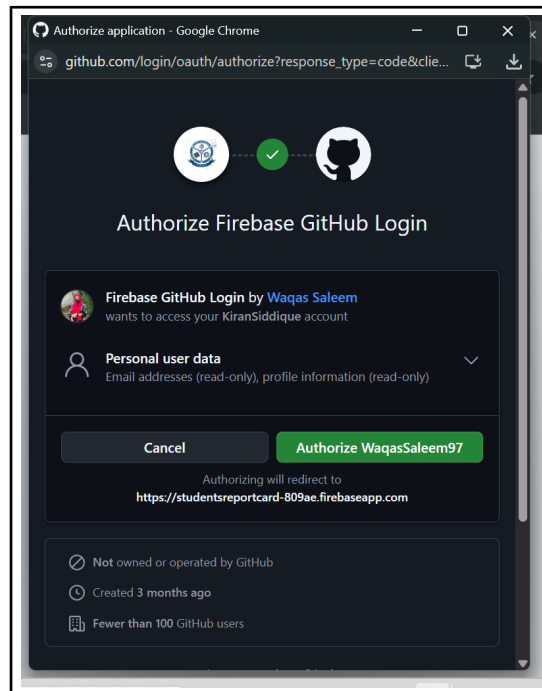


Figure: Student Marks Portal sign-in using Continue with GitHub (GitHub authorization step).

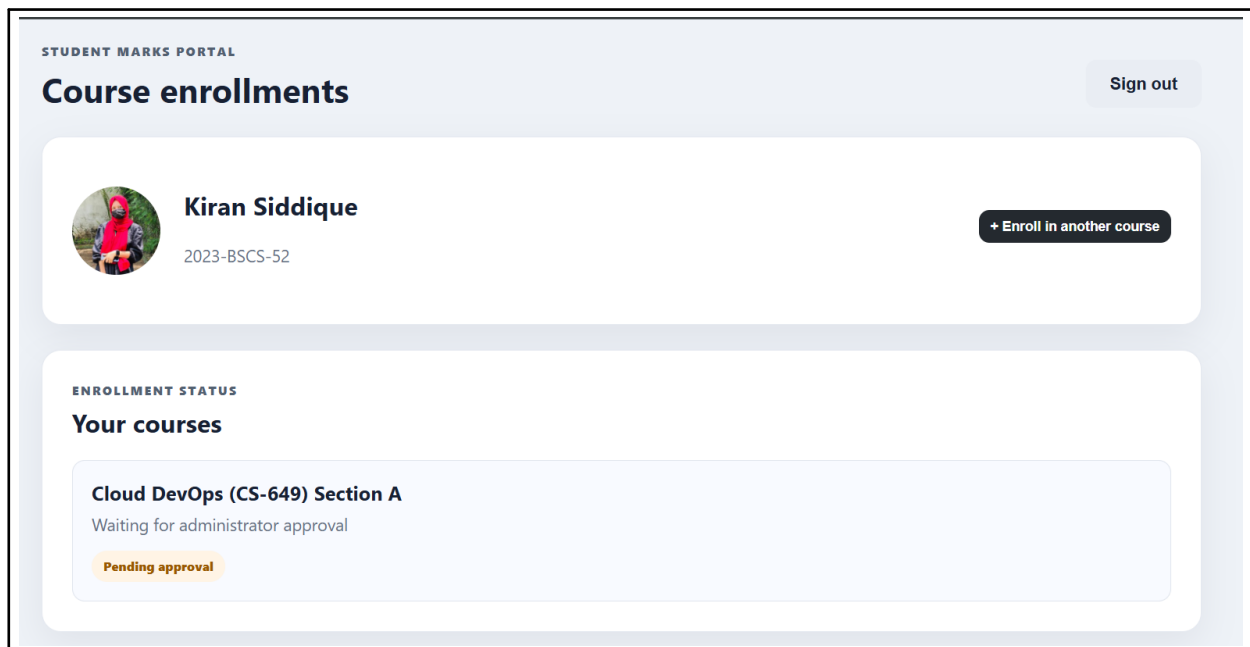


Figure: GitHub profile information (name and avatar) shown inside the portal.



STUDENT MARKS PORTAL

Course enrollments

Sign out



Kiran Siddique
2023-BSCS-52

+ Enroll in another course

ENROLLMENT STATUS

Your courses

Cloud DevOps (CS-649) Section A
Waiting for administrator approval

Pending approval

Figure: Submitted enrollment showing registration number and Pending approval status.



Task 3: Download VMware Workstation Pro and Ubuntu Server

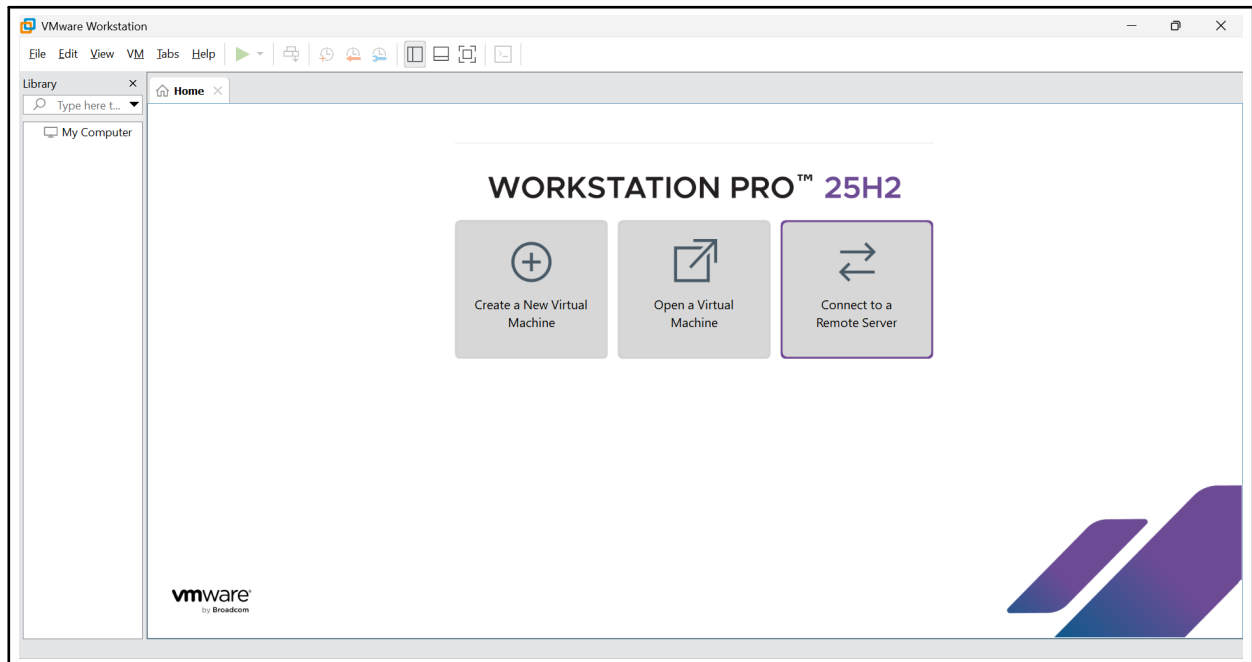


Figure: VMware Workstation Pro installed and running on Windows.

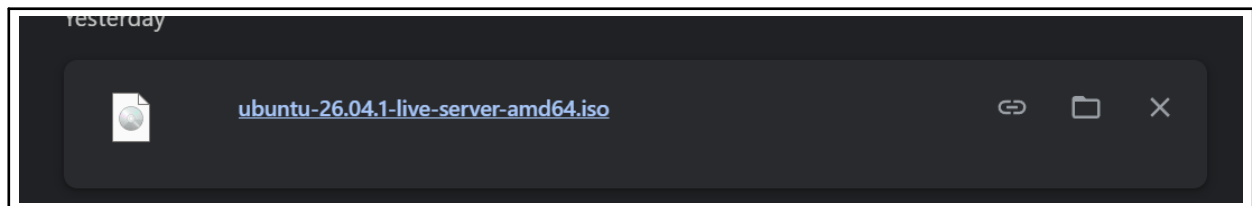


Figure: Downloaded Ubuntu Server LTS ISO file name.

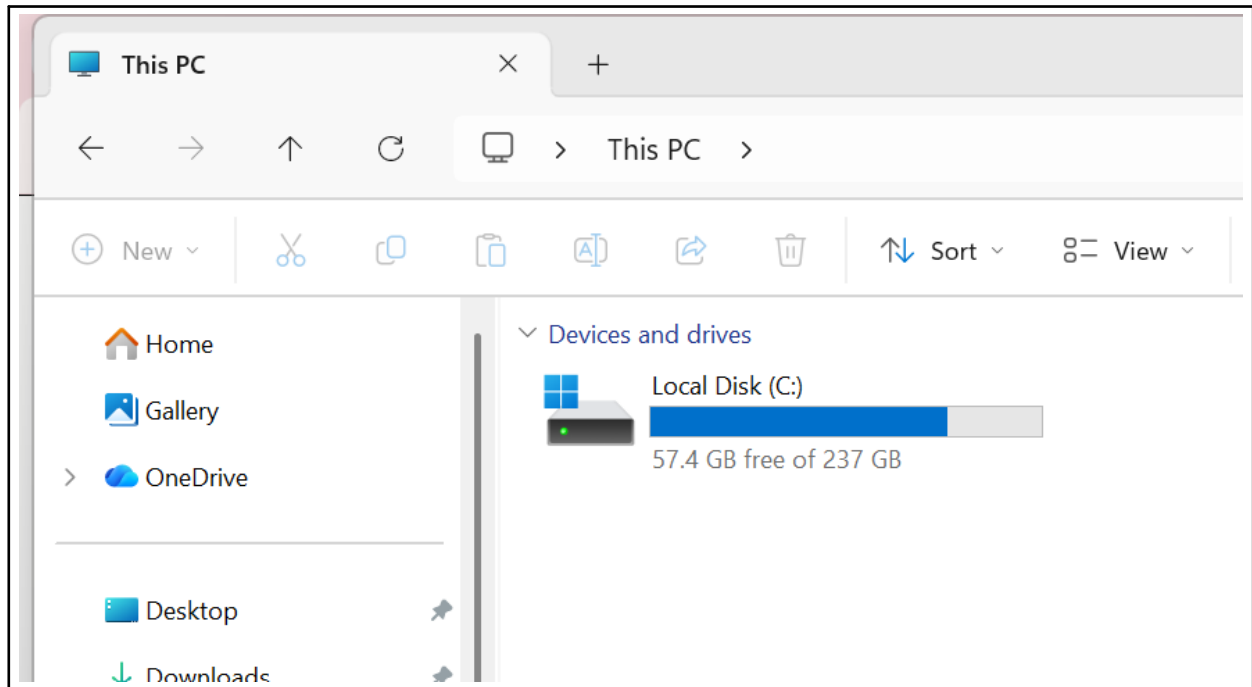


Figure: Available free storage on the host computer (at least 20 GB).



Task 4: Create the Ubuntu Server Virtual Machine

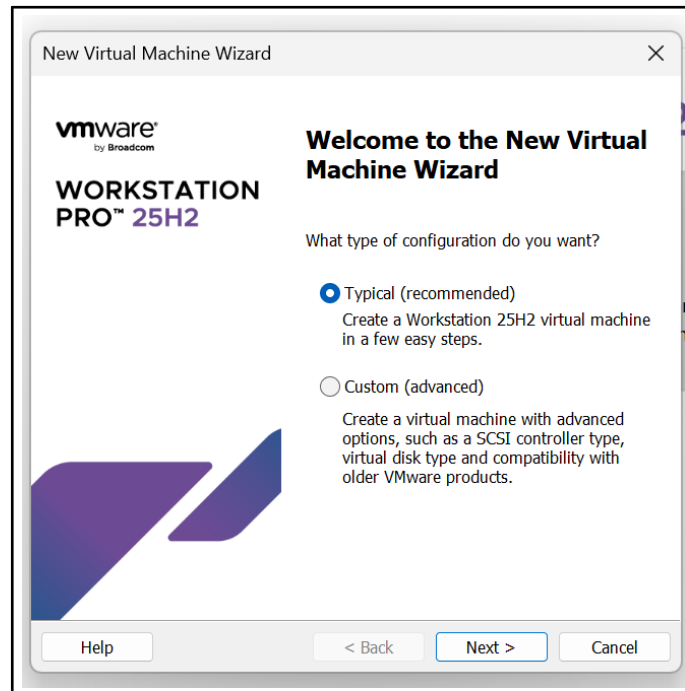


Figure: New Virtual Machine Wizard opened in VMware Workstation Pro.



Figure: Typical (recommended) configuration selected.

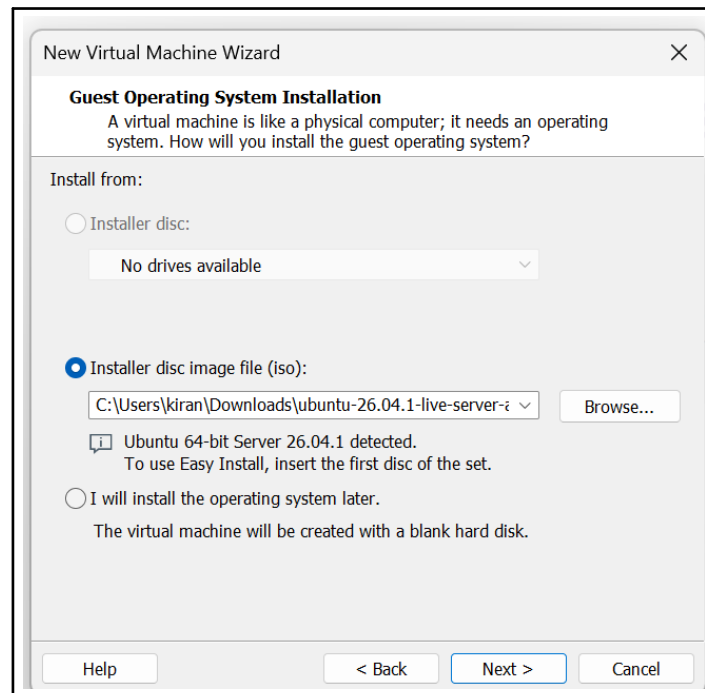


Figure: Installer disc image file (ISO) selected with the Ubuntu Server ISO path.

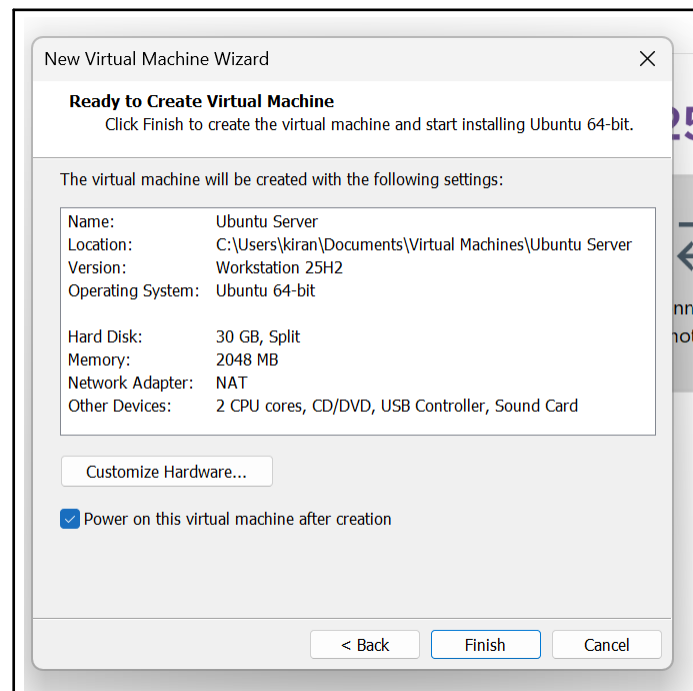


Figure: Final virtual machine configuration summary (30 GB disk).

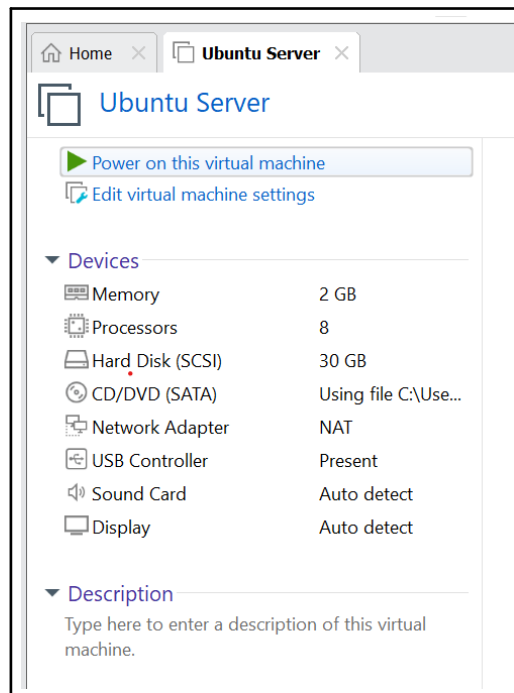


Figure: Virtual machine ready to be powered on.



Task 5: Choose the Installation Language

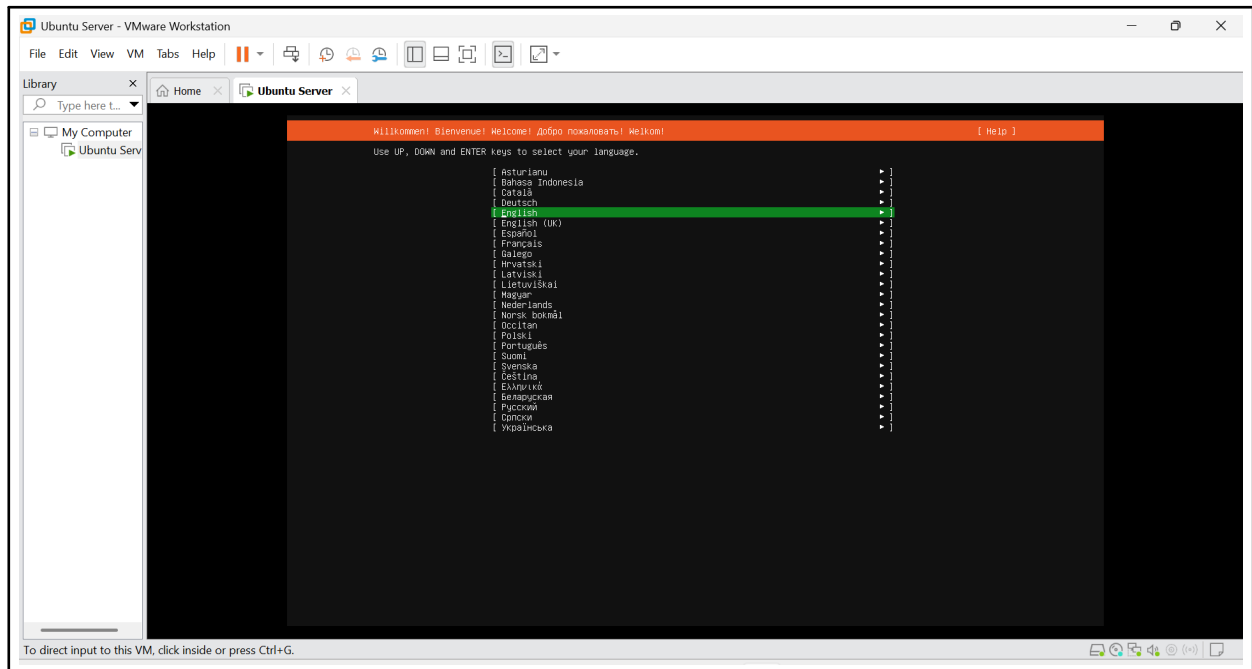


Figure: Installation language selected in the Ubuntu Server installer.



Task 6: Configure the Keyboard Layout

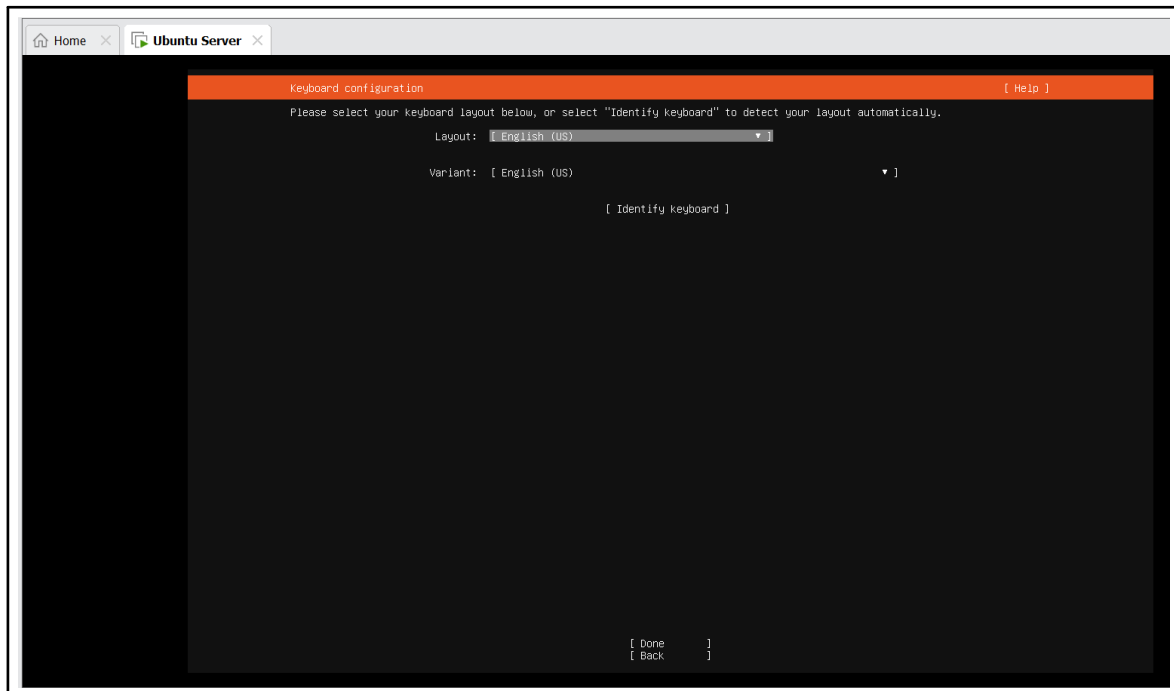


Figure: Keyboard layout selected.

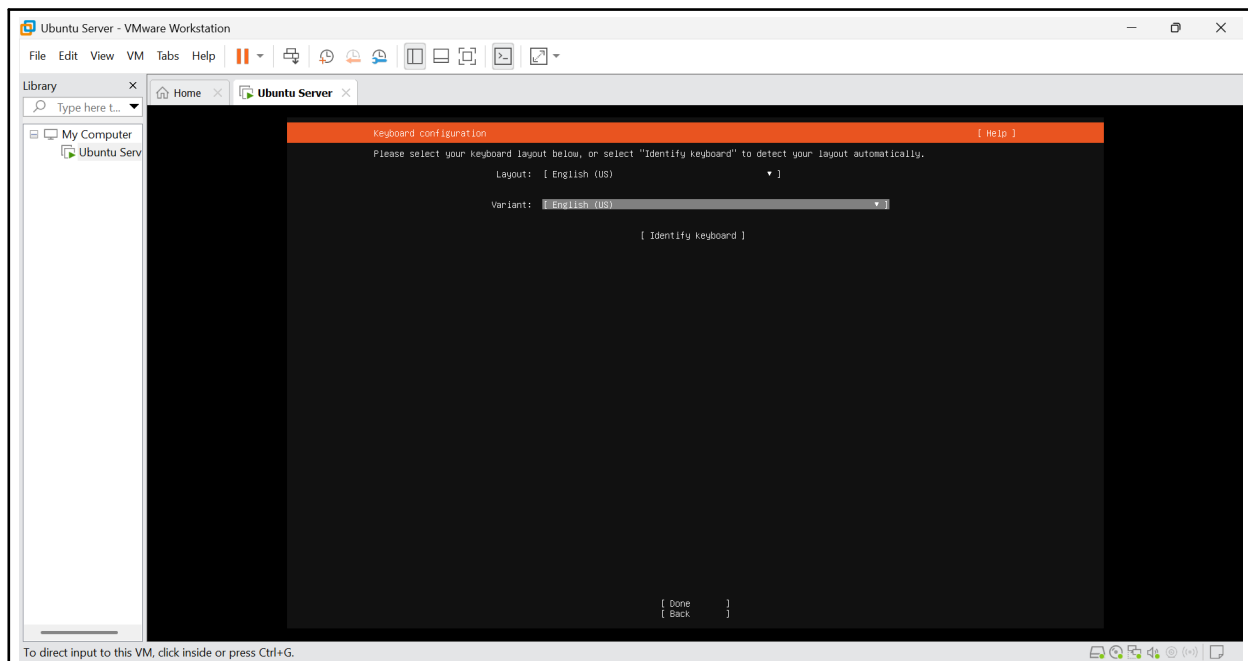


Figure: Keyboard variant selected.



Task 7: Choose the Ubuntu Server Installation Type

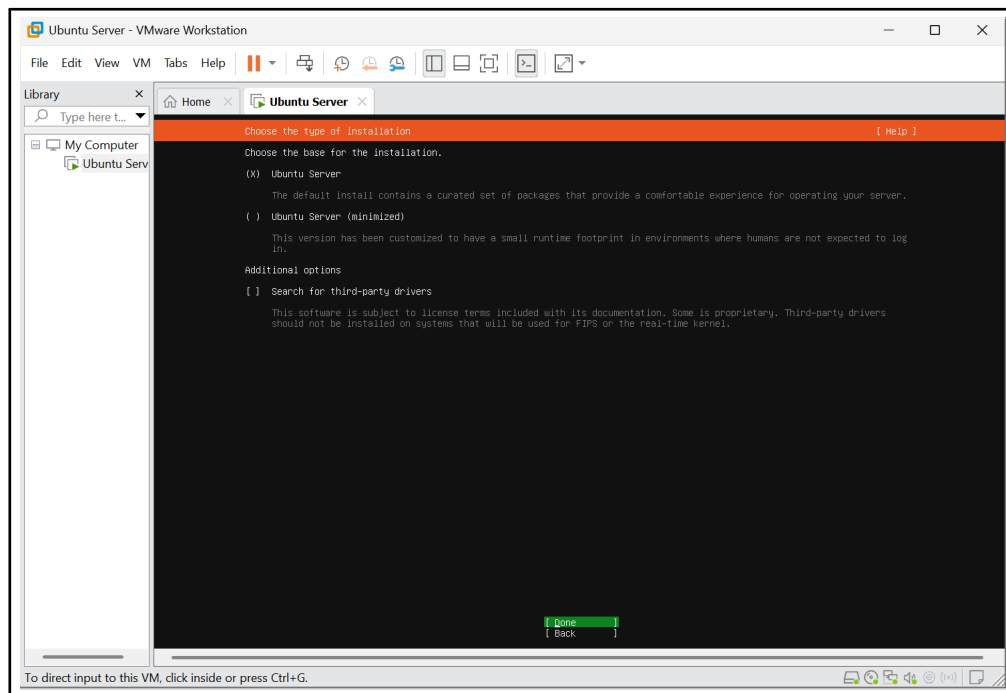


Figure: Normal Ubuntu Server installation type selected.



Task 8: Configure the Network

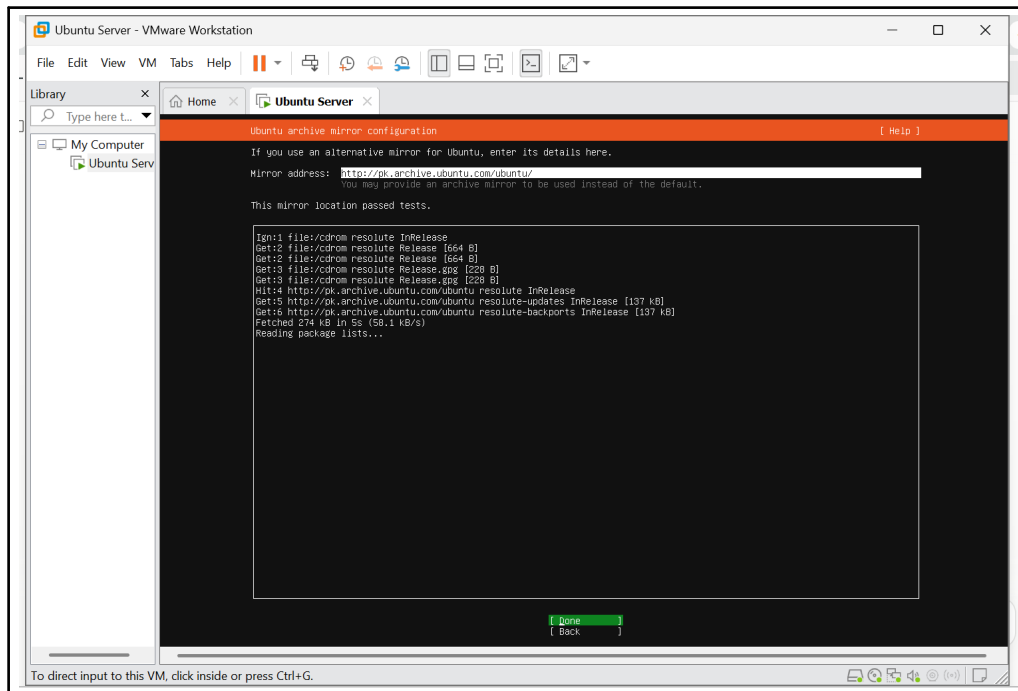


Figure: Archive mirror configuration with the mirror location test passed.



Task 9: Configure Storage

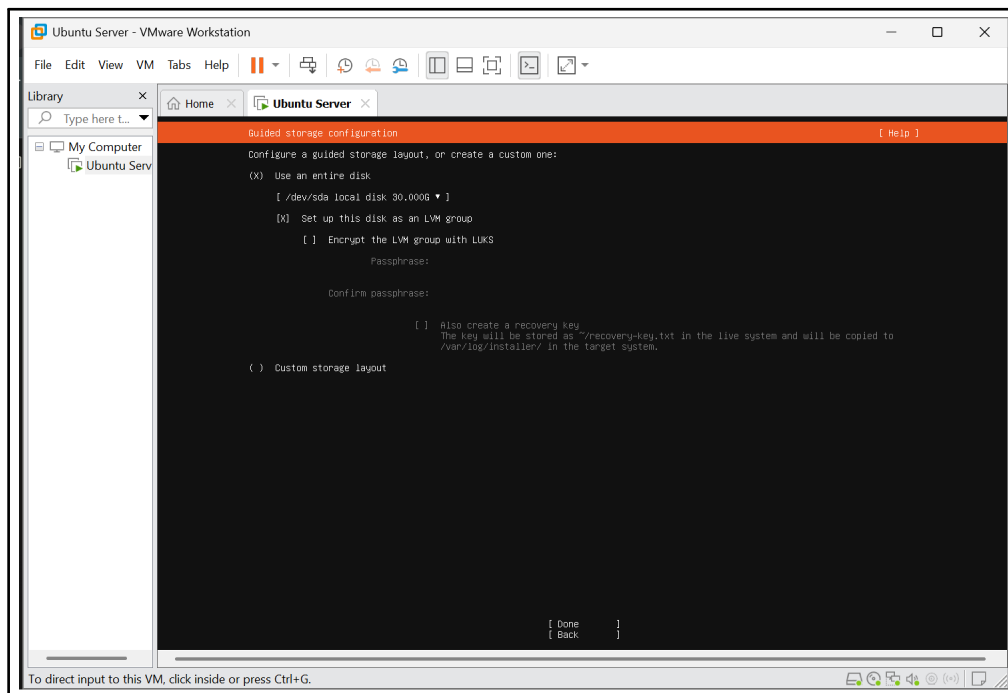


Figure: Guided storage option selected to use the VMware virtual disk.

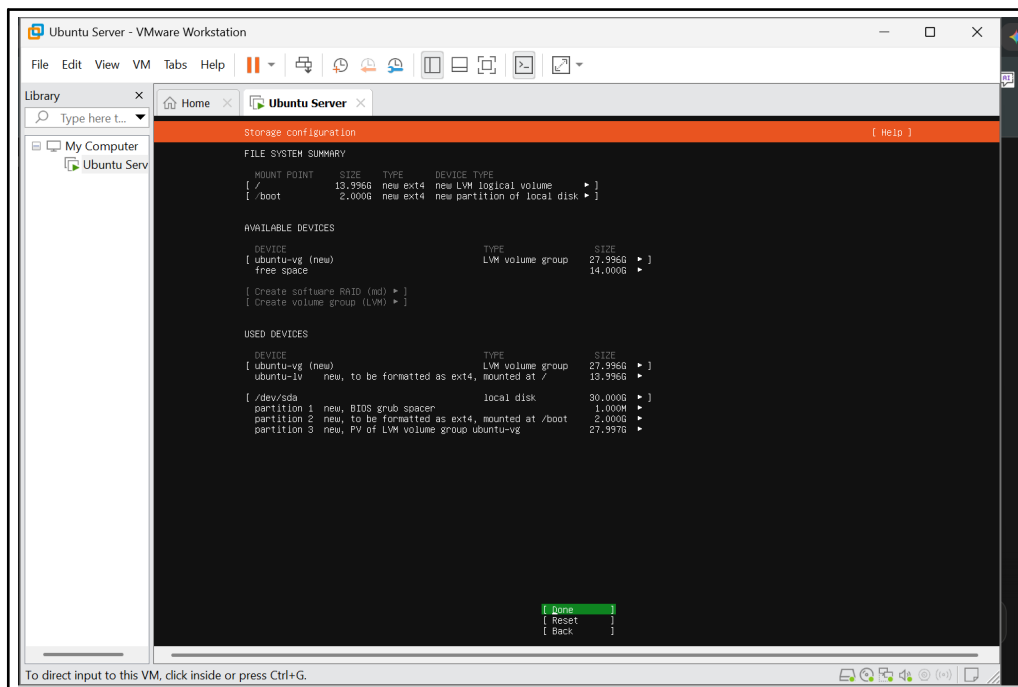


Figure: Storage configuration reviewed before continuing.



Task 10: Select the Installation Disk

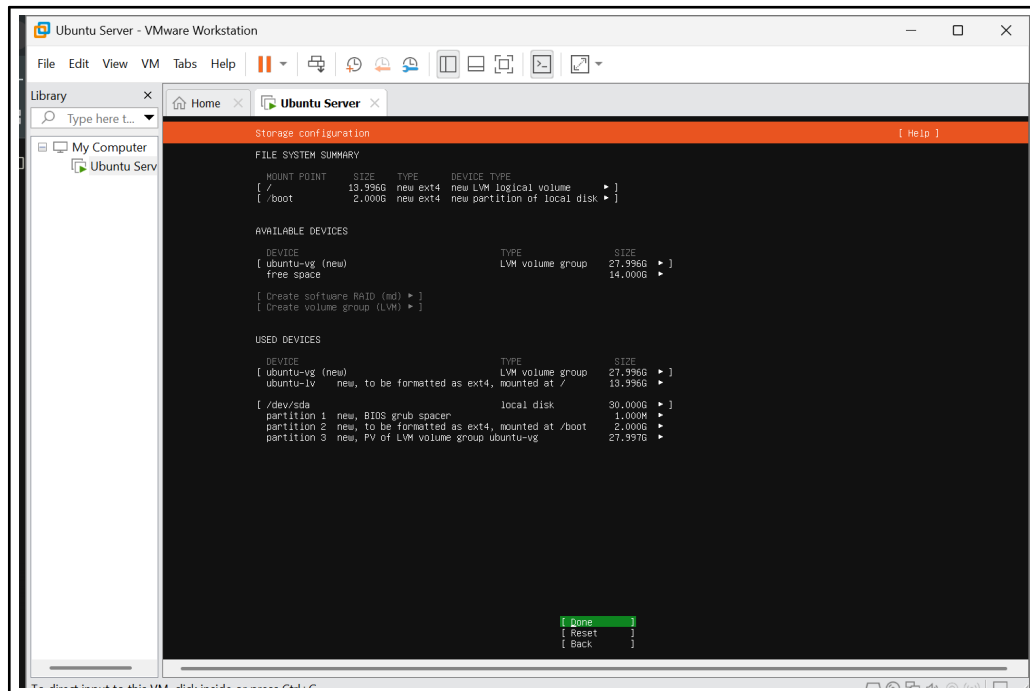


Figure: VMware virtual disk selected for installation.

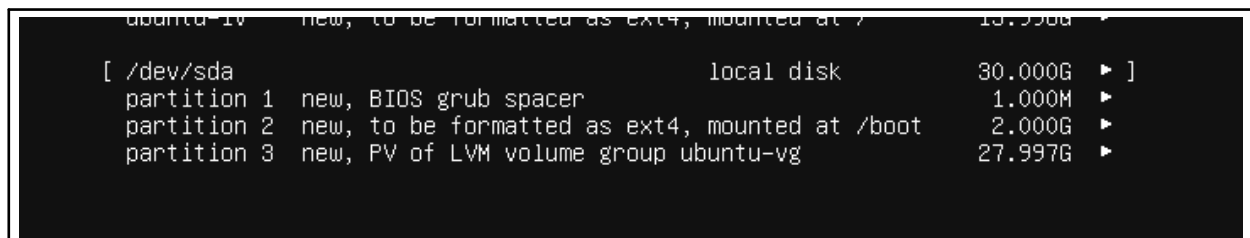


Figure: Disk name and size (30 GB) confirming the correct virtual disk.



Task 11: Review the Partitions

```
Storage configuration

FILE SYSTEM SUMMARY

MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /             13.996G new ext4 new LVM logical volume ► ]
[ /boot         2.000G new ext4 new partition of local disk ► ]

AVAILABLE DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                     LVM volume group                   27.996G ► ]
free space                             14.000G ► ]

[ Create software RAID (md) ► ]
[ Create volume group (LVM) ► ]

USED DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                     LVM volume group                   27.996G ► ]
ubuntu-lv    new, to be formatted as ext4, mounted at / 13.996G ► ]

[ /dev/sda                             local disk                         30.000G ► ]
partition 1  new, BIOS grub spacer                               1.000M ► ]
partition 2  new, to be formatted as ext4, mounted at /boot 2.000G ► ]
partition 3  new, PV of LVM volume group ubuntu-vg           27.997G ► ]
```

Figure: Partition layout proposed on /dev/sda (VMware virtual disk).



Task 12: Confirm the Storage Changes

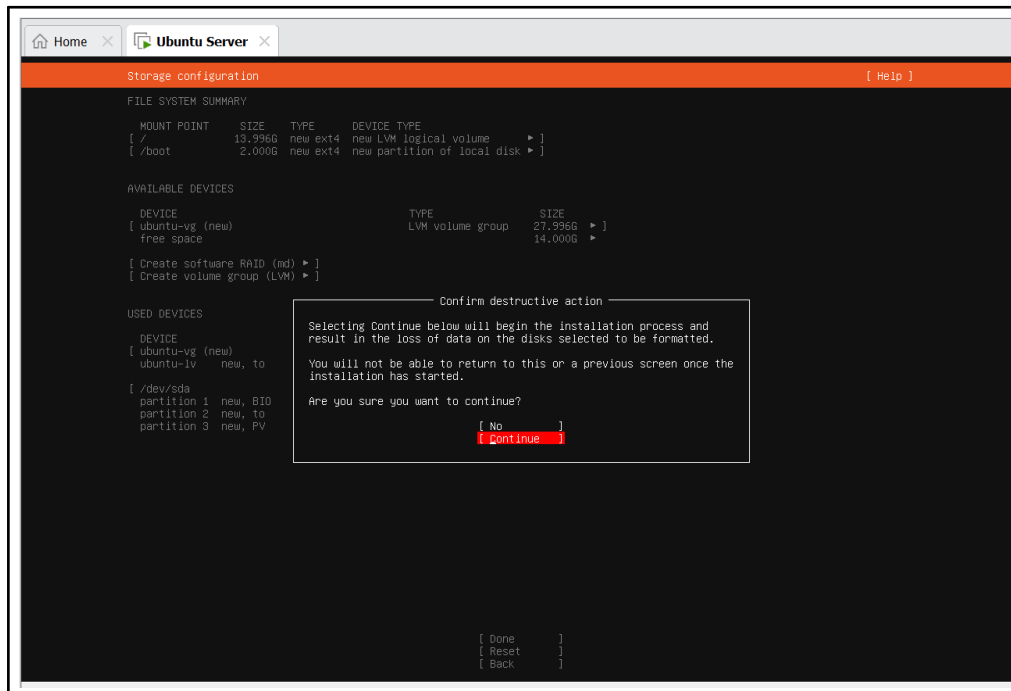


Figure: Destructive-action warning shown by the installer.

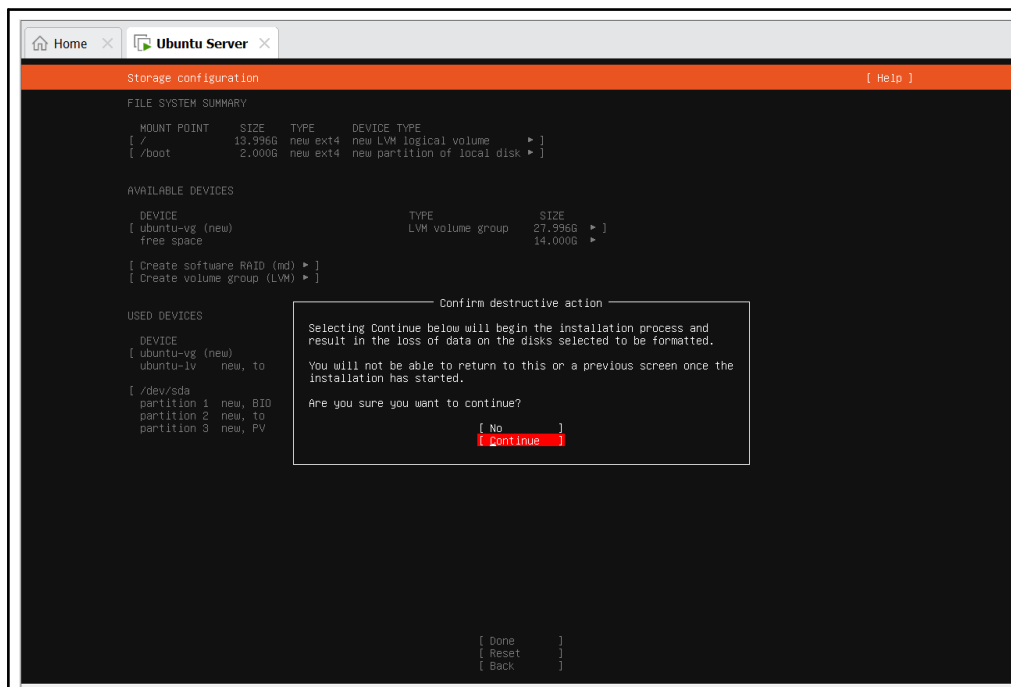


Figure: Storage changes confirmed with the virtual disk selected.



Task 13: Configure the Ubuntu Profile

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: xiransiddique

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: xiransiddique

Choose a password: *****

Confirm your password: *****

[Done]

Figure: Complete actual name entered in the Your name field.

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: xiransiddique

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: xiransiddique

Choose a password: *****

Confirm your password: *****

[Done]

Figure: Server name set to ubuntu.



Home × Ubuntu Server ×

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: kiraniddique

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: kiraniddique

Choose a password: *****

Confirm your password: *****

[Done]

Figure: Ubuntu username set to the exact GitHub username.

Home × Ubuntu Server ×

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: kiraniddique

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: kiraniddique

Choose a password: *****

Confirm your password: *****

[Done]

Figure: Completed profile configuration page (password hidden).

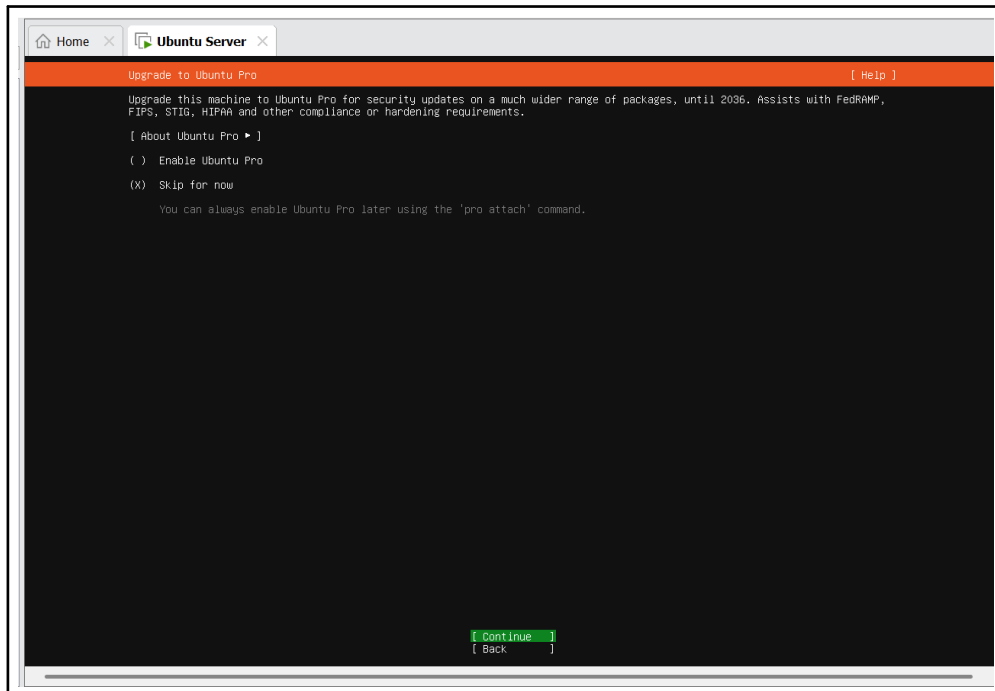


Figure: Upgrade to Ubuntu Pro screen with Skip for now selected.

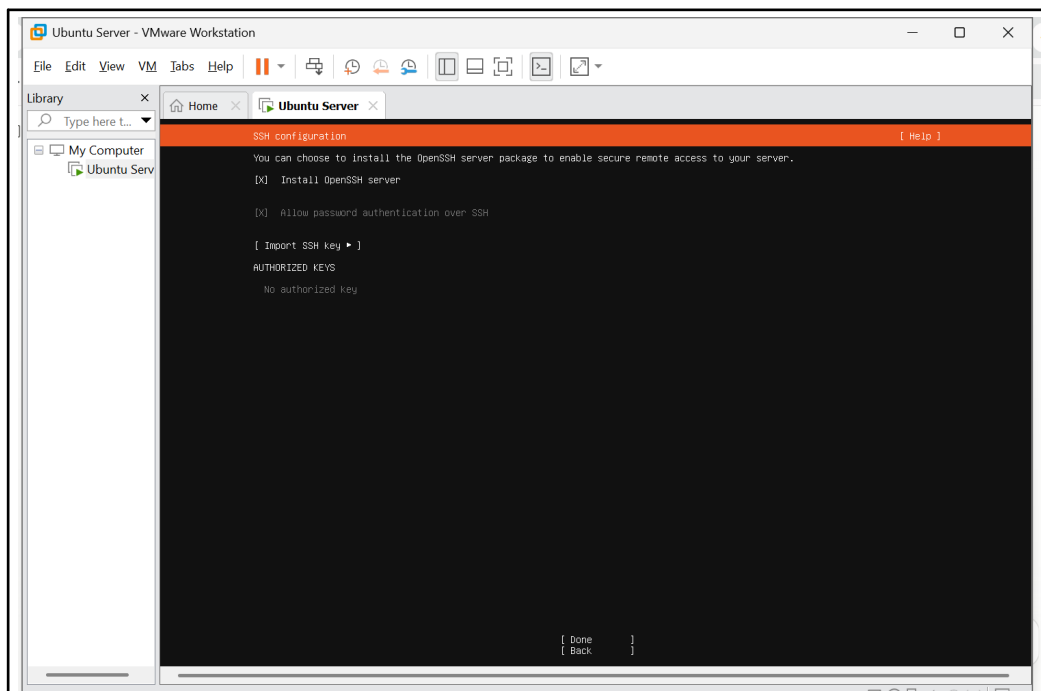


Figure: SSH configuration with Install OpenSSH server selected.



Task 14: Complete the Software Installation

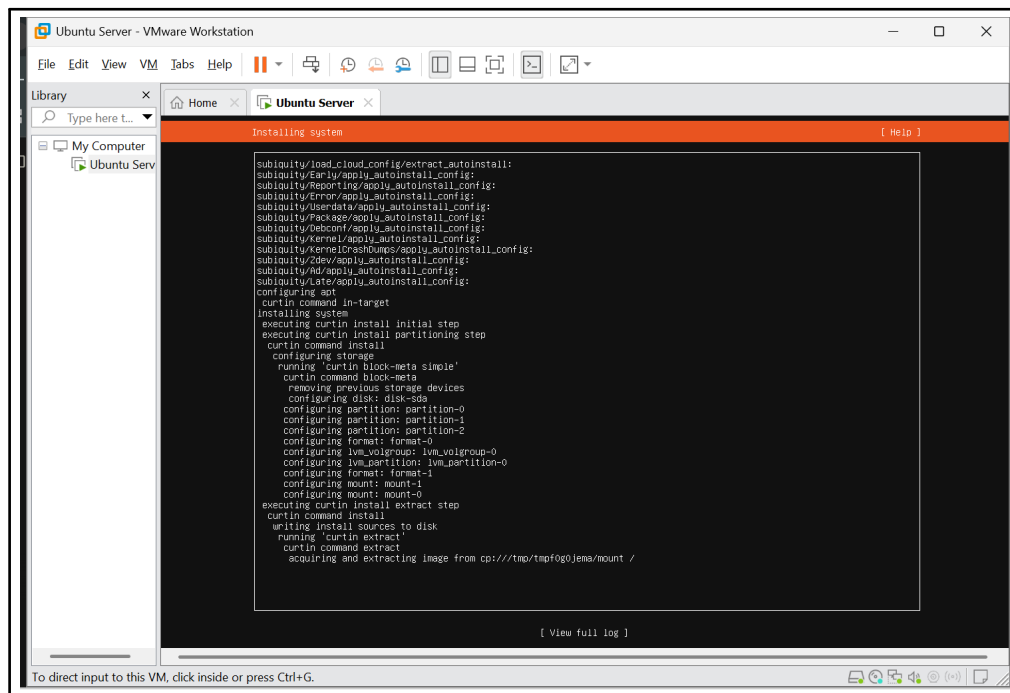


Figure: Installation progress while Ubuntu Server installs the required software.



Task 15: Reboot and Verify the Ubuntu Identity

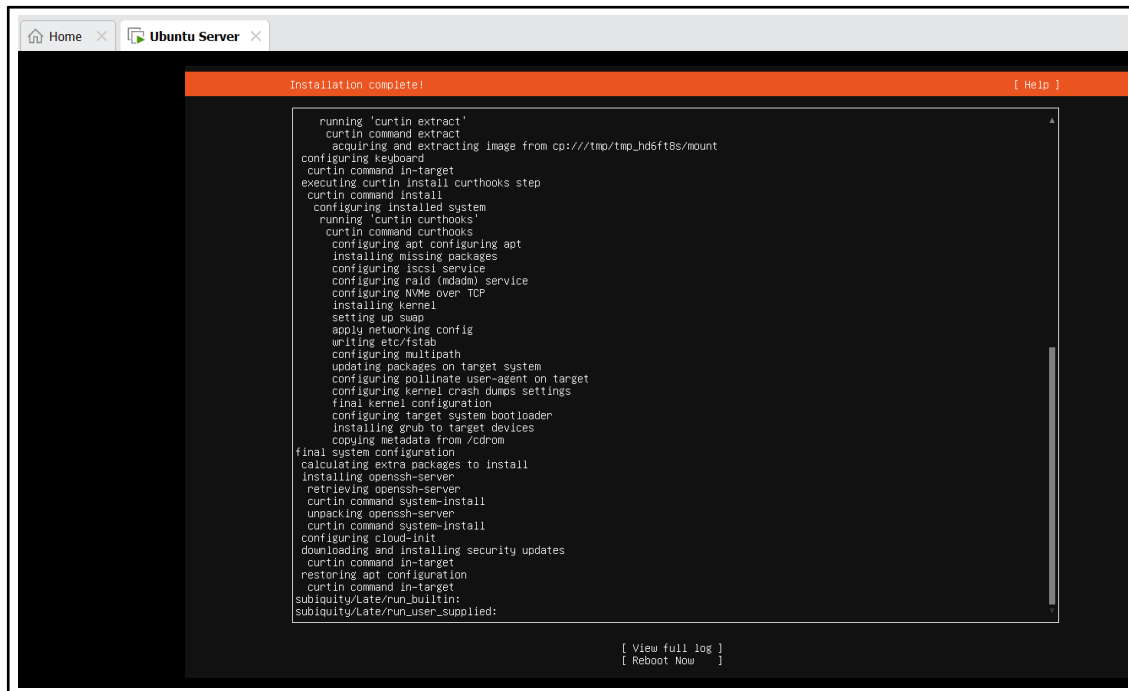


Figure: Installation complete screen with the Reboot Now option.

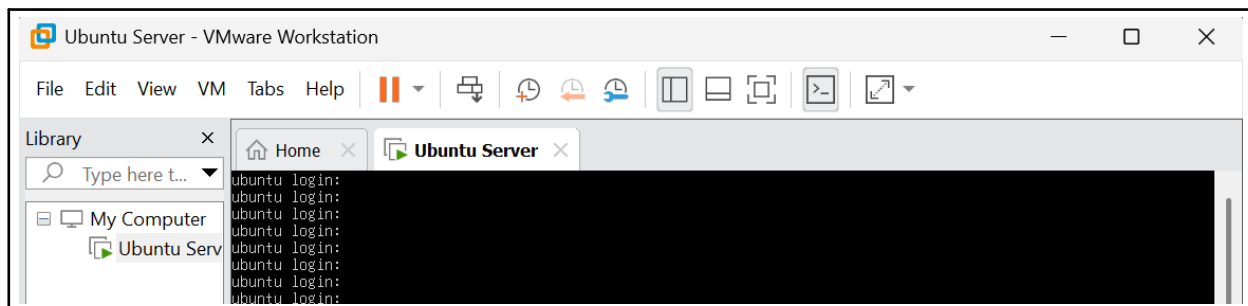


Figure: Ubuntu Server login screen after reboot.

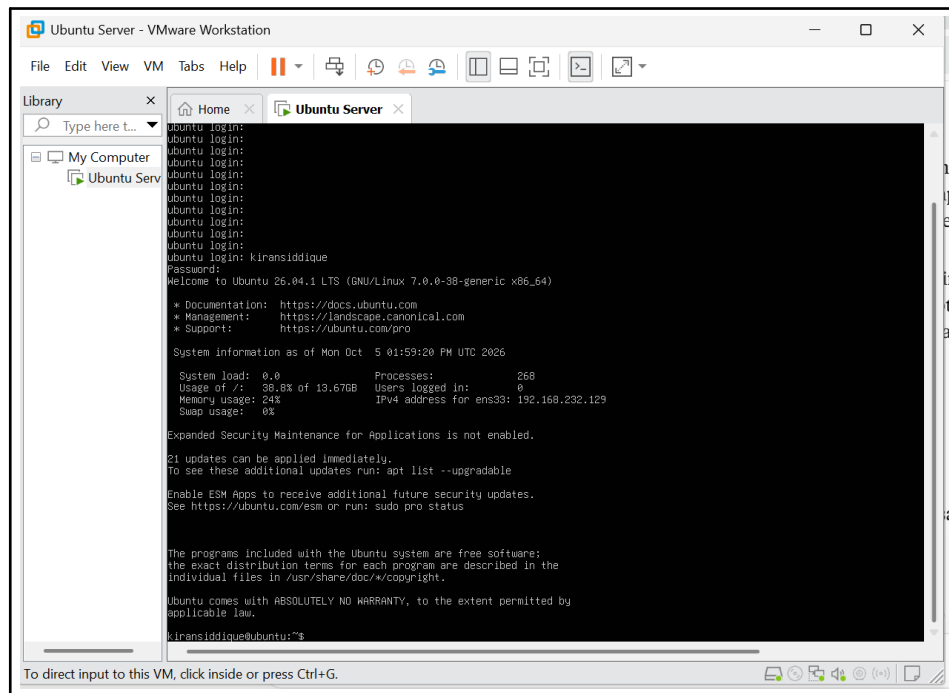


Figure: Successful terminal login with the configured username.



Figure: Prompt kiransiddique@ubuntu:~\$ verifying the required identity.



Task 16: Find the Ubuntu Server IP Address

```
kiransiddique@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:c2:af:bc brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altnamex enx000c29c2afbc
    inet 192.168.232.129/24 metric 100 brd 192.168.232.255 scope global dynamic ens33
        valid_lft 1255sec preferred_lft 1255sec
    inet6 fe80::20c:29ff:fec2:afbc/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
kiransiddique@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:c2:af:bc brd ff:ff:ff:ff:ff:ff
    altnamex enp2s1
```

Figure: The `ip addr` command and its output inside Ubuntu Server.

```
kiransiddique@ubuntu:~$
kiransiddique@ubuntu:~$ ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    altnamex enp2s1
    altnamex enx000c29c2afbc
    inet 192.168.232.129/24 metric 100 brd 192.168.232.255 scope global dynamic ens33
        valid_lft 1040sec preferred_lft 1040sec
kiransiddique@ubuntu:~$
```

/M, click inside or press Ctrl+G.

Figure: Active interface IPv4 address (192.168.232.129) with the prompt visible.



Task 17: Connect to Ubuntu Server from Windows

```
kiransiddique@ubuntu: ~  
Windows PowerShell  
Copyright (C) Microsoft Corporation. All rights reserved.  
  
PS C:\WINDOWS\system32> ssh kiransiddique@192.168.232.129  
The authenticity of host '192.168.232.129 (192.168.232.129)' is
```

Figure: SSH command run from Windows PowerShell.

```
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.232.129' (ED25519) to the list of known hosts  
kiransiddique@192.168.232.129's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Mon Oct  5 02:06:29 PM UTC 2026  
  
System load:  0.05          Processes:            267  
Usage of /:   38.8% of 13.67GB Users logged in:       0  
Memory usage: 25%          IPv4 address for ens33: 192.168.232.129  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
21 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status
```

Figure: Server fingerprint accepted by typing yes.

```
Warning: Permanently added '192.168.232.129' (ED25519) to the list of known hosts.  
kiransiddique@192.168.232.129's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)
```

Figure: Successful SSH login session (password not shown).



Figure: Remote prompt showing kiransiddique@ubuntu.